Merge Binary Trees

AlgoExpert · Medium · Binary Trees

The problem

- Overlay two trees; overlapping nodes sum their values.
- Mutates and returns tree1.

Intuition

- Overlay the trees node by node: add values where both exist, otherwise keep whichever exists.

Brute force vs optimal

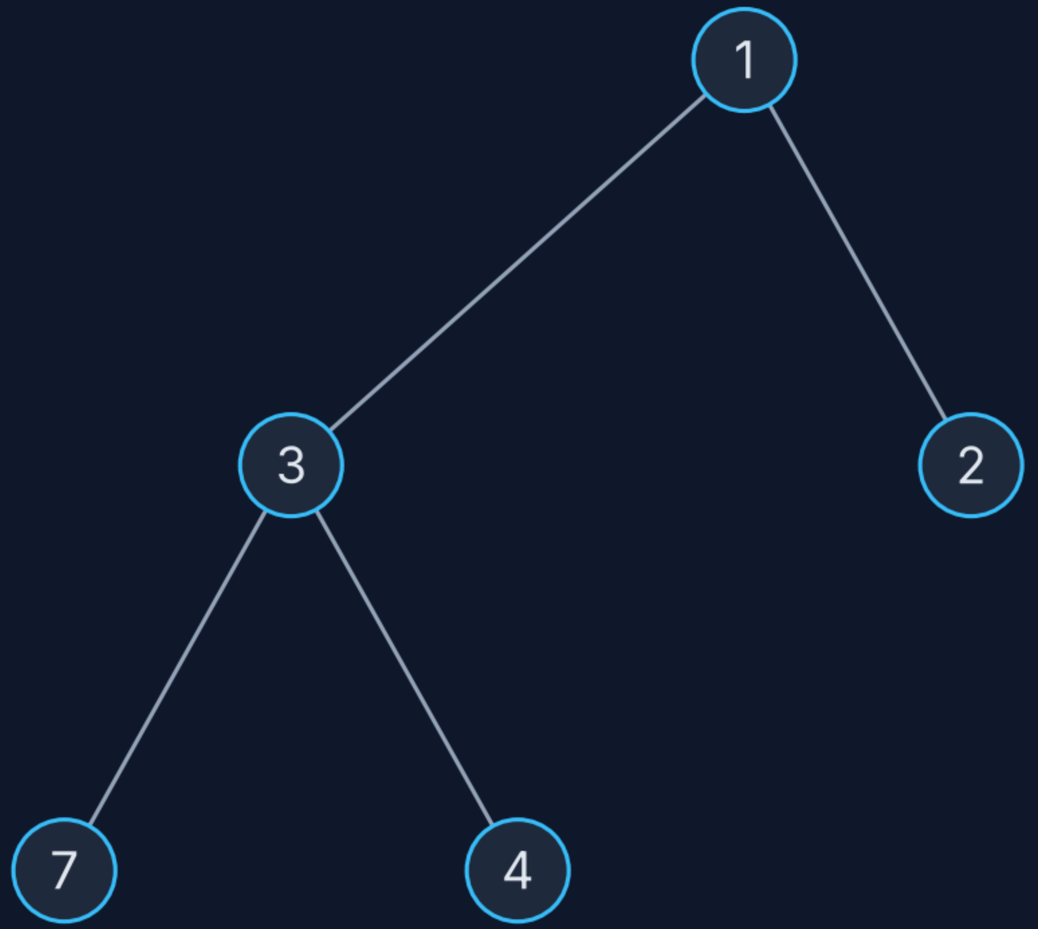
- Build a brand-new tree → O(n + m) extra space
- Merge into tree 1 in place → O(min(n, m)) time
- Reuse subtrees where only one side exists

Approach

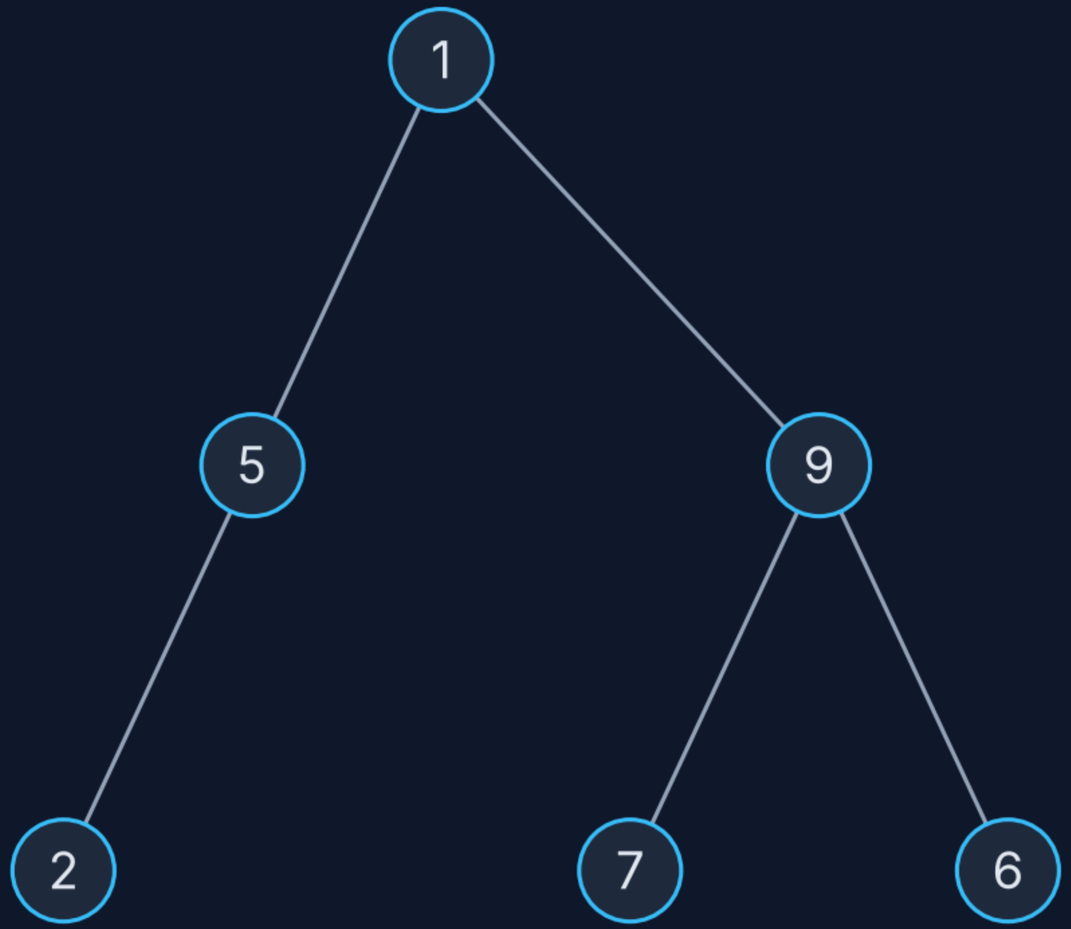
- 1. If one tree is null, return the other
- 2. Add the values
- 3. Merge left children and right children

Worked example

tree 1

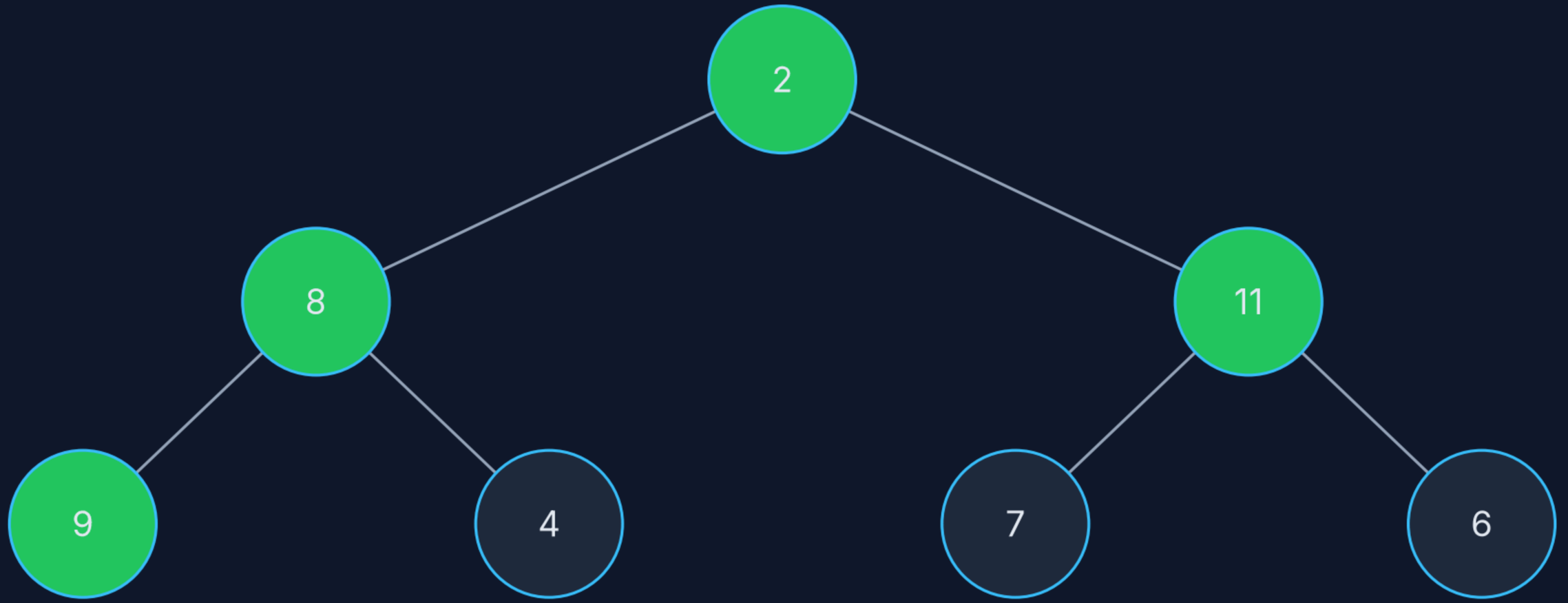


tree 2



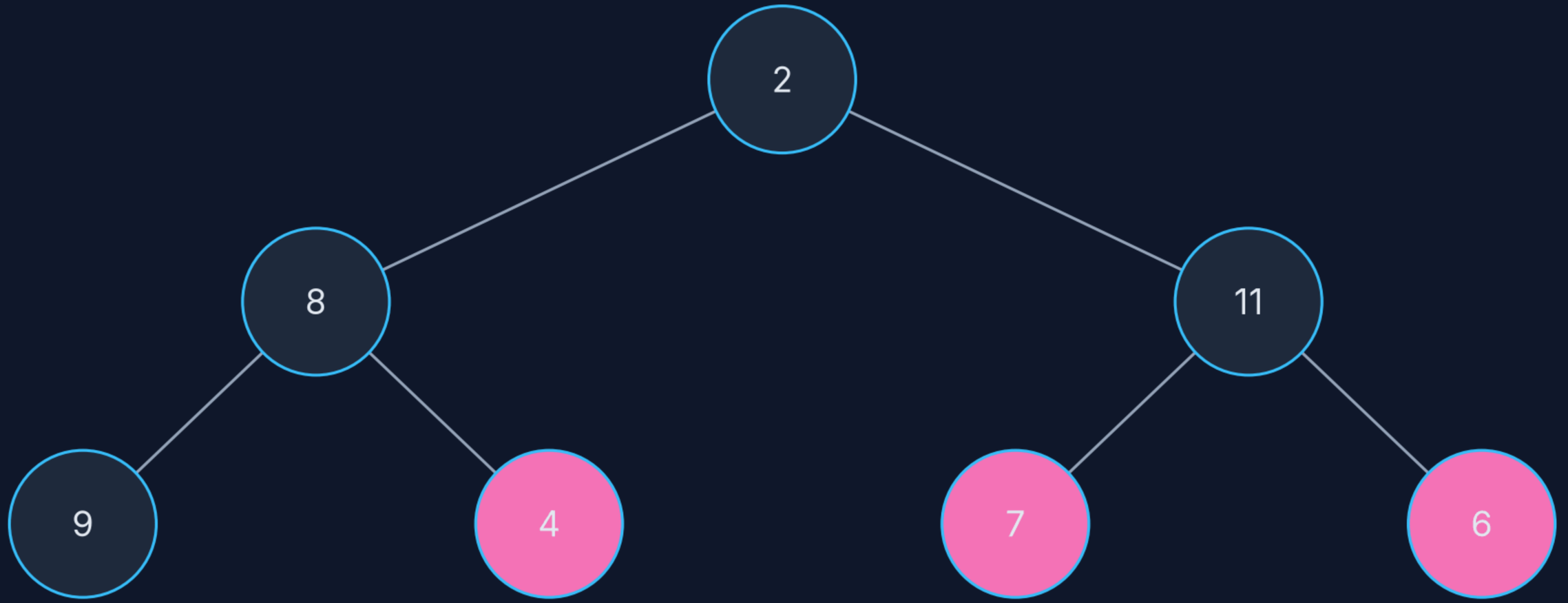
Two trees to overlay

Worked example



Overlapping nodes summed

Worked example



Missing on one side → reuse the other subtree

Java solution

```java
public class MergeBinaryTrees {
    public static BinaryTree mergeBinaryTrees(BinaryTree a, BinaryTree b) {
        if (a == null) return b;
        if (b == null) return a;
        a.value += b.value;
        a.left = mergeBinaryTrees(a.left, b.left);
        a.right = mergeBinaryTrees(a.right, b.right);
        return a;
    }
}
```

Complexity

- Time: O(n) of the smaller overlap
- Space: O(h)

Edge cases & pitfalls

- Either tree may be null
- Mutates tree 1 (mention it)
- Reused subtrees are shared

Interview tips & follow-ups

- Ask if inputs may be modified
- Iterative with a stack of pairs

Recap

- Overlay the trees node by node: add values where both exist, otherwise keep whichever exists.
- Complexity: O(n) of the smaller overlap